

Table level Operations with AWS CLI

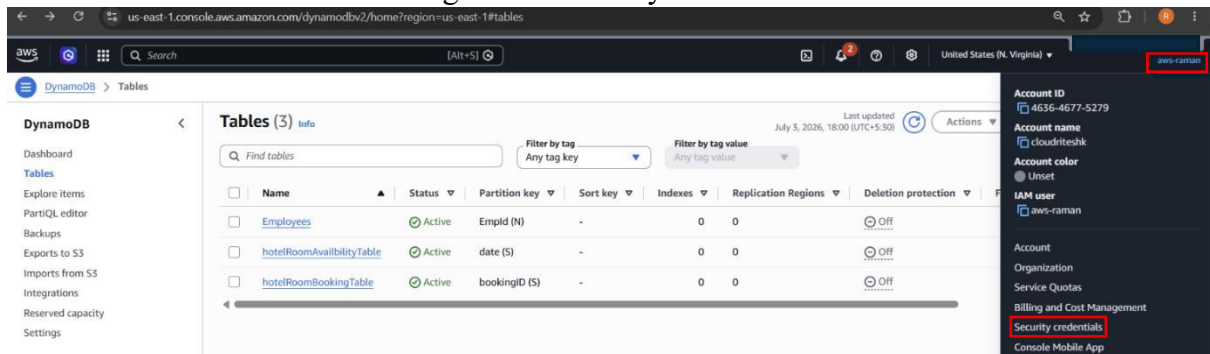
To Begin with the Lab

Summary of the Lab

This lab demonstrates DynamoDB table-level operations using the AWS CLI. It covers listing tables, describing table metadata, creating a new table by defining attributes, primary keys, and provisioned throughput, checking the table status until it becomes active, and verifying the created tables using the **list-tables** command.

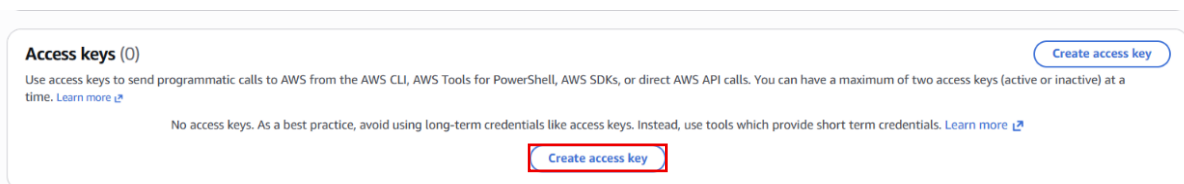
Lab Step

- Installing AWS Command Line Interface (CLI)
- Go to the AWS Account then go to the security credential in AWS account



Step 1: Configure AWS CLI (if not already)

- Go to the Access key in Security
- Click on the Create access key
- Create Access Key for aws configure
- aws configure is Terminal command
- I will use this command go the following



- Go to the Command prompt or VS code Terminal
- Use the **aws configure** command
- Put AWS Access key ID , AWS Secret Access Key and Default Region Name

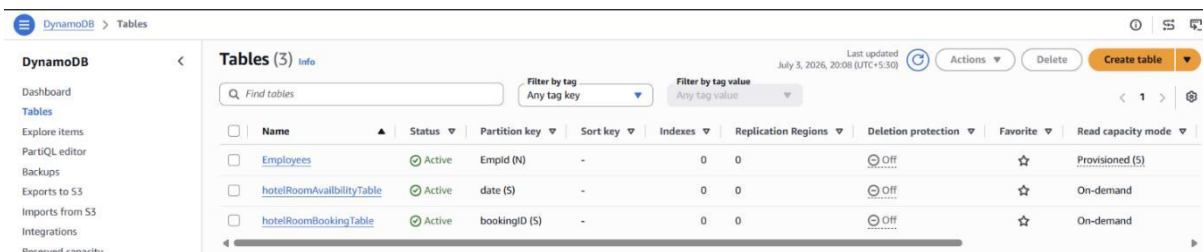
```
Windows PowerShell
Windows PowerShell 5.1
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\rrath> aws configure
AWS Access Key ID [*****72YS]:
AWS Secret Access Key [*****dgYb]:
Default region name [us-east-1]:
```

Step 2: List Existing DynamoDB Tables

- Show only three table and create fourth table to the terminal
- I will create table of student of Name



The screenshot shows the AWS DynamoDB console interface. On the left is a navigation menu with options like Dashboard, Tables, Explore items, PartiQL editor, Backups, Exports to S3, Imports from S3, and Integrations. The main area is titled 'Tables (3)' and contains a table listing existing DynamoDB tables. The table has columns for Name, Status, Partition key, Sort key, Indexes, Replication Regions, Deletion protection, Favorite, and Read capacity mode. Three tables are listed: 'Employees' (Provisioned (5)), 'hotelRoomAvailabilityTable' (On-demand), and 'hotelRoomBookingTable' (On-demand).

Name	Status	Partition key	Sort key	Indexes	Replication Regions	Deletion protection	Favorite	Read capacity mode
Employees	Active	EmpId (N)	-	0	0	Off	☆	Provisioned (5)
hotelRoomAvailabilityTable	Active	date (S)	-	0	0	Off	☆	On-demand
hotelRoomBookingTable	Active	bookingID (S)	-	0	0	Off	☆	On-demand

Step 3: Create a New Table

- Go the terminal
- I have use of some command of table
- **aws dynamodb create-table** this command use of Create a table
- **--table-name** this command use of **Identify the table**
- **--attribute-definitions** this command use of AttributeName and AttributeType
- **Key -Schema** command use of AttributeName and keyType
- **--billing-mode** command use for payment
- Name of Table is **Student** ,Attributename is **StudentID** , AttributeType is **S(string)**
- Paste the command for **Create table**

```
aws dynamodb create-table --table-name Students --attribute-definitions
AttributeName=StudentId,AttributeType=S --key-schema
AttributeName=StudentId,KeyType=HASH --billing-mode PAY_PER_REQUEST -region
ap-south-1
```

```
PS C:\Users\rrath> aws dynamodb create-table --table-name Students --attribute-definitions AttributeName=StudentId,AttributeType=S --
key-schema AttributeName=StudentId,KeyType=HASH --billing-mode PAY_PER_REQUEST --region ap-south-1

{
  "TableDescription": {
    "AttributeDefinitions": [
      {
        "AttributeName": "StudentId",
        "AttributeType": "S"
      }
    ],
    "TableName": "Students",
    "KeySchema": [
      {
        "AttributeName": "StudentId"

        "KeyType": "HASH"

      }
    ],
    "TableStatus": "CREATING",
    "CreationDateTime": "2026-07-03T20:27:33.297000+05:30",
    "ProvisionedThroughput": {
      "NumberOfDecreasesToday": 0,
      "ReadCapacityUnits": 0,
      "WriteCapacityUnits": 0
    }
  }
}
```

Step 4: Check Table Status

- Go to the DynamoDB in AWS Account
- Click on the table section
- Check the table is created

The screenshot shows the AWS DynamoDB console interface. On the left is a sidebar with navigation options like Dashboard, Tables, Explore items, etc. The main area displays a list of tables under the heading 'Tables (4)'. The tables listed are Employees, hotelRoomAvailabilityTable, hotelRoomBookingTable, and Students. The 'Students' table is highlighted with a red rectangular box. The table's status is 'Active', its partition key is 'StudentId (S)', and its sort key is '-'. It has 0 indexes and 0 replication regions. Deletion protection is 'Off', and its read capacity mode is 'On-demand'.

Name	Status	Partition key	Sort key	Indexes	Replication Regions	Deletion protection	Favorite	Read capacity m
Employees	Active	Emplid (N)	-	0	0	Off	☆	Provisioned (5)
hotelRoomAvailabilityTable	Active	date (S)	-	0	0	Off	☆	On-demand
hotelRoomBookingTable	Active	bookingID (S)	-	0	0	Off	☆	On-demand
Students	Active	Studentid (S)	-	0	0	Off	☆	On-demand

Write Operations- item level Oproations

Step 7 Create second Table

- Go to the terminal
- Create new table **td_notes_test** to the same way

```
aws dynamodb create-table --table-name td_notes_test --attribute-definitions
AttributeName=StudentId,AttributeType=S --key-schema
AttributeName=StudentId,KeyType=HASH --billing-mode PAY_PER_REQUEST -region ap-
south-1
```

DynamoDB > Tables

Tables (5) info

Filter by tag: Any tag key

Filter by tag value: Any tag value

☐	Name	Status	Partition key	Sort key	Indexes	Replication Regions	Deletion protection	Favorite	Read capacity mode
☐	Employees	Active	EmpId (N)	-	0	0	Off	☆	Provisioned (5)
☐	hotelRoomAvailabilityTable	Active	date (S)	-	0	0	Off	☆	On-demand
☐	hotelRoomBookingTable	Active	bookingID (S)	-	0	0	Off	☆	On-demand
☐	Students	Active	user_id (S)	timestamp (N)	0	0	Off	☆	On-demand
☐	td_notes_test	Active	user_id (S)	timestamp (N)	0	0	Off	☆	On-demand

Step 8: Delete Command

- Go to the Terminal
- Use the **delete table command**
- `aws dynamodb delete-table --table-name td_notes_test`
- Table is **deleting successful**

```
PS C:\Users\rrath> aws dynamodb delete-table --table-name td_notes_test
{
  "TableDescription": {
    "TableName": "td_notes_test",
    "TableStatus": "DELETING",
    "ProvisionedThroughput": {
      "NumberOfDecreasesToday": 0,
      "ReadCapacityUnits": 0,
      "WriteCapacityUnits": 0
    },
    "TableSizeBytes": 0,
    "ItemCount": 0,
    "TableArn": "arn:aws:dynamodb:us-east-1:463646775279:table/td_notes_test",
    "TableId": "a1843fed-e2bb-427b-9ccc-ae39e6be9b87",
    "BillingModeSummary": {
      "BillingMode": "PAY_PER_REQUEST",
      "LastUpdateToPayPerRequestDateTime": "2026-07-04T12:05:18.953000+05:30"
    },
    "DeletionProtectionEnabled": false
  }
}
```

DynamoDB > Tables

Tables (4) info

Filter by tag: Any tag key

Filter by tag value: Any tag value

☐	Name	Status	Partition key	Sort key	Indexes	Replication Regions	Deletion protection	Favorite	Read capacity m
☐	Employees	Active	EmpId (N)	-	0	0	Off	☆	Provisioned (5)
☐	hotelRoomAvailabilityTable	Active	date (S)	-	0	0	Off	☆	On-demand
☐	hotelRoomBookingTable	Active	bookingID (S)	-	0	0	Off	☆	On-demand
☐	Students	Active	Studentid (S)	-	0	0	Off	☆	On-demand

Step 9: Create item in Table

- Go to the VS code create Folder, name of **CLI**
- Create a new file in CLI folder, in **item.json**
- Run these Code in **item.json** file

```
{
  "user_id": {"S": "1234abcd"},
  "timestamp": {"N": "1525012300E"},
  "content": {"S": "This is a sample note"},
  "title": {"S": "my title"}
}
```

- Go to the VS code Terminal

- Run this Command
- `aws dynamodb put-item --table-name Students --item file://C:\CLI\item.json`

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\CLI> aws dynamodb put-item --table-name Students --item file://C:\CLI\item.json
PS C:\CLI>

```

Step 10: Check Item Status

- Go to the DynamoDB in AWS Account
- Click on the **Explore items**
- Choose the Student table
- Check the item in Students Table

The screenshot shows the AWS DynamoDB console interface. On the left, the 'Explore Items' link is highlighted with a red box. The main panel shows the 'Students' table selected. Below the 'Scan or query items' section, a green status bar indicates 'Completed - Items returned: 2 - Items scanned: 2 - Efficiency: 100% - RCUs consumed: 2'. The table view shows two items. The first item has a 'user_id' of '1234abcd' (highlighted with a red box), a 'timestamp' of '1525012927', and a 'title' of 'my title'.

- Go to Terminal, **Create new item**

```

{"user_id": {"S": "2224jgfgj"},
"timestamp": {"N": "7577358"}}

```

- Again run this command
- `aws dynamodb put-item --table-name Students --item file://C:\CLI\item.json` • Second item created successfully

The screenshot shows the AWS DynamoDB console interface. The main panel shows the 'Students' table selected. Below the 'Scan or query items' section, a green status bar indicates 'Completed - Items returned: 2 - Items scanned: 2 - Efficiency: 100% - RCUs consumed: 2'. The table view shows two items. The first item has a 'user_id' of '1234abcd', a 'timestamp' of '1525012927', and a 'title' of 'my title'. The second item has a 'user_id' of '2224jgfgj' (highlighted with a red box) and a 'timestamp' of '7577358'.

- Go to the VS code, **Open CLI Folder**
- Create files in CLI folder
- Key.json, attribute-value.json, item.json and attribute-name.json

- **Key.json code is**

```
{
  "user_id": {"S": "2224jgfgj"},
  "timestamp": {"N": "7577358"}
}
```

- **attribute-value.json code is**

```
{
  "t": {"S": "My updated title"}
}
```

- **item.json code is**

```
{
  "user_id": {"S": "2224jgfgj"},
  "timestamp": {"N": "7577358"},
  "content": {"S": "This is a sample note"},
  "title": {"S": "my title"}
}
```

- **attribute-name.json code is**

```
{
  "#t": "title"
}
```

- Paste this Command

```
aws dynamodb update-item --table-name Students --key file://key.json --update-expression
"SET #t = :t" --expression-attribute-names file://attribute-names.json -expression-
attribute-values file://attribute-values.json
```

```
key.json
1
2 user_id: {"S": "2224jgfgj"},
3 timestamp: {"N": "7577358"}
4

attribute-value.json
1
2 My updated title
3

item.json
1 {
2   "user_id": {"S": "2224jgfgj"},
3   "timestamp": {"N": "7577358"},
4   "content": {"S": "This is a sample note"},
5   "title": {"S": "my title"}
6 }

attribute-name.json
1
2 "#t": "title"
3
```

```
PS C:\CLI> aws dynamodb update-item --table-name Studentd --key file://key.json --update-expression "SET #t = :t" --expression-attribute-names file://attribute-names.json --expression-attribute-values file://attribute-values.json
```

- Go to the dynamodb in AWS Account then go **explore item**
- Choose students table and check User_id 2224jgfgj is Updated

Table: Students - Items returned (2)

Scan started on July 04, 2026, 13:00:15

	user_id (String)	timestamp (Number)	content	title
☐	2224jgfgj	7577358	My update ...	title
☐	1234abcd	1525012927	This is a sa...	my title